Solutions to Review Questions for Lecture 3

2025-1-30

1. Sorting the list gives the values $2, 6, 9, 10, 50$. It follows that 20% of the values are less than or equal to 2 and 40% are less than or equal to 6. Hence 6 is the 23rd percentile based on the definition in class.
2. The value will be the mean of $3, 5, 4, 8$, which is $(3 + 5 + 4 + 8)/4 = 5$.
3. The value will be the median of $6, 1, -1, 0, 4$. Sorting the list gives $-1, 0, 1, 4, 6$. The value 1 will be the median, since 60% of the values are less than or equal to 1 and only 40% are less than or equal to 0.
4. The 25th percentile is 6 and the 75th percentile 10 since 80% of the values are less than or equal to 10 while only 60% of the values are less than or equal to 9. Hence the IQR is $10 - 6 = 4$. The median will be 9 since 60% of the values are less than or equal to 9 while only 40% are less than or equal to 6. The absolute values of the deviations from the median are $7, 3, 0, 1, 43$. Sorting this list gives $0, 1, 3, 7, 43$ with $MAD = 3$.
5. A middle class person looking to purchase a house might be interested in the median house price in a city, while a property tax assessor for the city might be more interested in the mean house price.

```
test = c(0.32, 0.72, 1.48, 1.50, -0.58, 0.46, -0.72, -0.31, 0.86, 0.15, 0.46, 1.87, 0.11,
         0.26, -0.37, -0.99, 1.15, 0.01, -1.75, -0.51, 1.75, 0.75, -0.69, 0.40, -1.30)
print(mean(test)) #Q6
```

```
## [1] 0.2012
```

```
print(median(test)) #Q7
```

```
## [1] 0.26
```

```
#R calculates SD+, and we need to adjust by multiplying by sqrt((n-1)/n) to get SD
n = length(test)
print(sd(test)*sqrt((n-1)/n)) #Q8
```

```
## [1] 0.9268282
```

```
print(sqrt(mean(test**2)-mean(test)**2)) #Q8 calculated directly
```

```
## [1] 0.9268282
```